

List of Items — Java Collections Demo

A small Java console project that demonstrates how to store, access, search, modify, and display a collection of Item objects using every major Java data structure: arrays, 2D arrays, List, ArrayList, LinkedList, Set, HashSet, and HashMap.

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Project Purpose

This project was built as a learning exercise to compare how the same set of operations (Access, Search, Modify, Display) behave across different Java collection types. Each section in ListOfItems.java is self-contained and prints its own labeled output block to the console.

Project Structure

File	Description
Item.java	The model class representing a single item
ListOfItems.java	Main class demonstrating all data structures

The Item Class

The Item class represents one item. Each item has:

Field	Description
ID	Unique identifier for the item
Name	The item's name
Price	The item's price
Category	The item's category

- The constructor creates the item.
- Getters get the data.
- Setters change the name, price, and category.
- The display() method prints the item information.

Main Class

The program creates five items: Laptop, Phone, Book, Chair, and Table. Then it shows different Java data structures.

1. Array

Stores items in one row. We can access, search, change, and display items.

2. Two-Dimensional Array

Stores items in rows and columns. We access items using two indexes.

3. List

Stores items in order. We can get, search, change, and display items.

4. ArrayList

Stores items and can grow in size. We can add and modify items.

5. LinkedList

Stores items in linked order. We can access and modify items.

6. Set

Stores unique items. We can search and modify items.

7. HashSet

A Set that does not keep a fixed order. We can add, search, and modify items.

8. HashMap

Stores items using a key and value. Here, the ID is the key and the Item is the value.

In Short

The program shows how to store, access, search, modify, and display items using different Java data structures.

Sample Output

Screenshots of the console output for each data structure section:

| EXECUTING TESTS (METHOD C11) @ DISOLIVERS

===== 1D ARRAY =====

===== access =====

First item: Laptop

Second item catogrey: Electronics

=====

Found: Book

ID: 1

Name: Laptop

Price: 25000.0

Category: Electronics

=====

ID: 2

Name: Phone

Price: 14000.0

Category: Electronics

=====

ID: 3

Name: electronics

Price: 300.0

Category: Education

=====

ID: 4

Name: Chair

Price: 1200.0

Category: education

=====

=====

=====

ID: 5

Name: Table

Price: 2500.0

Category: Furniture

=====

=====

===== 2D ARRAY =====

===== 2D ARRAY Access =====

Item at [0][1]: Phone

Item at [1][1]: education

Item at [2][0]: 2500.0

Furniture: Table

ID: 1

Name: Laptop

Price: 25000.0

Category: Electronics

=====

ID: 2

Name: Phone

Price: 14000.0

Category: Electronics

=====

ID: 3

Name: Java Programming Book

Price: 300.0

Category: Education

=====

```
=====
ID: 4
Name: Chair
Price: 1600.0
Category: education
=====
```

```
ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
===== LIST =====
Item at index 2: Java Programming Book
Item at index 3: Chair
Found Laptop
ID: 1
Name: Laptop
Price: 24000.0
Category: Electronics
=====
```

```
ID: 2
Name: Phone
Price: 14000.0
Category: Electronics
=====
```

```
ID: 3
Name: Java Programming Book
Price: 300.0
Category: Education
=====
```

```
ID: 4
Name: Chair
Price: 1600.0
Category: education
=====
```

```
ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
===== ARRAYLIST =====
Item at index 1: Phone
Expensive item: Laptop
Expensive item: Phone
ID: 1
Name: Laptop
Price: 24000.0
Category: Electronics
=====
ID: 2
Name: Phone
Price: 14000.0
Category: Electronics
=====
```

```
=====
ID: 3
Name: Java Programming Book
Price: 350.0
Category: Education
=====

ID: 4
Name: Chair
Price: 1600.0
Category: education
=====

ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====

===== LINKEDLIST =====
First item: Laptop
Education item: Java Programming Book
ID: 1
Name: Laptop
Price: 24000.0
Category: Electronics
=====

ID: 2
Name: Phone
Price: 14000.0
Category: Electronics
=====

=====
ID: 3
Name: Java Programming Book
Price: 350.0
Category: Education
=====

ID: 4
Name: Chair
Price: 1300.0
Category: education
=====

ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====

===== SET =====
Found item: Chair
ID: 1
Name: Laptop
Price: 23000.0
Category: Electronics
=====

ID: 6
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
=====
ID: 2
Name: Phone
Price: 14000.0
Category: Electronics
=====
```

```
ID: 3
Name: Java Programming Book
Price: 350.0
Category: Education
=====
```

```
ID: 4
Name: Chair
Price: 1300.0
Category: education
=====
```

```
ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
===== HASHSET =====
Found Phone
ID: 1
Name: Laptop
Price: 23000.0
Category: Electronics
=====
```

```
=====
ID: 6
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
ID: 2
Name: Phone
Price: 14500.0
Category: Electronics
=====
```

```
ID: 3
Name: Java Programming Book
Price: 350.0
Category: Education
=====
```

```
ID: 4
Name: Chair
Price: 1300.0
Category: education
=====
```

```
ID: 5
Name: Table
Price: 2500.0
Category: Furniture
=====
```

```
===== HASHMAP =====
Item with ID 3: Java Programming Book
Item with ID 4 exists.
ID: 1
Name: Laptop
Price: 23000.0
Category: Electronics
=====
ID: 2
Name: Phone
Price: 14500.0
Category: Electronics
=====
ID: 3
Name: Java Programming Book
Price: 350.0
Category: Education
=====
ID: 4
Name: Chair
Price: 1300.0
Category: education
=====
ID: 5
Name: Table
Price: 2700.0
Category: Furniture
=====
```